1) a)

$$\frac{1}{\sqrt{a^2-x^2}} = \frac{1}{\sqrt{a^2-a^2\sin^2\theta}} = \frac{1}{\sqrt{a^2(1-\sin^2\theta)}} = \frac{1}{\sqrt{a^2\cos^2\theta}} = \frac{1}{a\cos\theta}$$

So use $x = a\sin\theta$

1) b)

$$\frac{1}{\sqrt{a^2+x^2}} = \frac{1}{\sqrt{a^2+a^2\sinh^2\theta}} = \frac{1}{\sqrt{a^2(1+\sinh^2\theta)}} = \frac{1}{\sqrt{a^2\cosh^2\theta}} = \frac{1}{a\cosh\theta}$$

So use $x = a\sinh\theta$

1) c)

$$\frac{1}{a^2-x^2} = \frac{1}{a^2-a^2\tanh^2\theta} = \frac{1}{a^2(1-\tanh^2\theta)} = \frac{1}{a^2\,\mathrm{sech}^2\theta}$$

So use $x = a\tanh\theta$

1) d)

$$\frac{1}{\sqrt{x^2-a^2}} = \frac{1}{\sqrt{a^2\cosh^2\theta-a^2}} = \frac{1}{\sqrt{a^2(\cosh^2\theta-1)}} = \frac{1}{\sqrt{a^2\sinh^2\theta}} = \frac{1}{a\sinh\theta}$$

So use $x = a\cosh\theta$

1) e)

$$\frac{1}{a^2+x^2} = \frac{1}{a^2+a^2\tan\theta} = \frac{1}{a^2(1+\tan^2\theta)} = \frac{1}{a^2\sec^2\theta}$$

So use $x = a\tan\theta$

2) a)

Let $9x^2 = 400\sinh^2 u \Rightarrow 3x = 20\sinh u \Rightarrow x = \frac{20}{3}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{20}{3}\cosh u \Rightarrow \mathrm{d}x = \frac{20}{3}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{9x^2+400}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{400\sinh^2 u+400}} \times \frac{20}{3}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{400(\sinh^2 u+1)}} \times \frac{20}{3}\cosh u\,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{3}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{1}{3}\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\mathrm{arsinh}\left(\frac{3}{20}x\right) + c$

$x = \frac{20}{3}\sinh u$ gives $\int \frac{1}{\sqrt{9x^2+400}}\mathrm{d}x = \frac{1}{3}\mathrm{arsinh}\left(\frac{3}{20}x\right) + c$

2) b)

Let $16x^2 = 9\sin^2 u \Rightarrow 4x = 3\sin u \Rightarrow x = \frac{3}{4}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{4}\cos u \Rightarrow \mathrm{d}x = \frac{3}{4}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{9-16x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9-9\sin^2 u}} \times \frac{3}{4}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(1-\sin^2 u)}} \times \frac{3}{4}\cos u\,\mathrm{d}u$

$= \frac{1}{4}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{4}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{4}\mathrm{d}u$

$= \frac{1}{4}u + c$

$= \frac{1}{4}\arcsin\left(\frac{4}{3}x\right) + c$

So $x = \frac{3}{4}\sin u$ gives $\int \frac{1}{\sqrt{9-16x^2}}\mathrm{d}x = \frac{1}{4}\arcsin\left(\frac{4}{3}x\right) + c$

2) c)

Let $x^2 = 36\tanh^2 u \Rightarrow x = 6\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 6\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = 6\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{36-x^2}\mathrm{d}x$

$= \int \frac{1}{36-36\tanh^2 u} \times 6\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{36(1-\tanh^2 u)} \times 6\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{6}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{1}{6}\mathrm{d}u$

$= \frac{1}{6}u + c$

$= \frac{1}{6}\operatorname{artanh}\left(\frac{1}{6}x\right) + c$

So $x = 6\tanh u$ gives $\int \frac{1}{36-x^2}\mathrm{d}x = \frac{1}{6}\operatorname{artanh}\left(\frac{1}{6}x\right) + c$

2) d)

Let $625x^2 = \sinh^2 u \Rightarrow 25x = \sinh u \Rightarrow x = \frac{1}{25}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{25}\cosh u \Rightarrow \mathrm{d}x = \frac{1}{25}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{625x^2+1}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{\sinh^2 u+1}} \times \frac{1}{25}\cosh u\,\mathrm{d}u$

$= \frac{1}{25}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{25}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{1}{25}\mathrm{d}u$

$= \frac{1}{25}u + c$

$= \frac{1}{25}\operatorname{arsinh} 25x + c$

So $x = \frac{1}{25}\sinh u$ gives $\int \frac{1}{\sqrt{625x^2+1}}\mathrm{d}x = \frac{1}{25}\operatorname{arsinh} 25x + c$

2) e)

Let $121x^2 = 16\cosh^2 u \Rightarrow 11x = 4\cosh u \Rightarrow x = \frac{4}{11}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{4}{11}\sinh u \Rightarrow \mathrm{d}x = \frac{4}{11}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{121x^2-16}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{16\cosh^2 u-16}} \times \frac{4}{11}\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{16(\cosh^2 u-1)}} \times \frac{4}{11}\sinh u\,\mathrm{d}u$

$= \frac{1}{11}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{1}{11}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int \frac{1}{11}\mathrm{d}u$

$= \frac{1}{11}u + c$

$= \frac{1}{11}\operatorname{arcosh}\left(\frac{11}{4}x\right) + c$

So $x = \frac{4}{11}\cosh u$ gives $\int \frac{1}{\sqrt{121x^2-16}}\mathrm{d}x = \frac{1}{11}\operatorname{arcosh}\left(\frac{11}{4}x\right) + c$

2) f)

Let $25x^2 = \tan^2 u \Rightarrow 5x = \tan u \Rightarrow x = \frac{1}{5}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{5}\sec^2 u \Rightarrow \mathrm{d}x = \frac{1}{5}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{1+25x^2}\mathrm{d}x$

$= \int \frac{1}{1+\tan^2 u} \times \frac{1}{5}\sec^2 u\,\mathrm{d}u$

$= \frac{1}{5}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{5}\mathrm{d}u$

$= \frac{1}{5}u + c$

$= \frac{1}{5}\arctan 5x + c$

So $x = \frac{1}{5}\tan u$ gives $\int \frac{1}{1+25x^2}\mathrm{d}x = \frac{1}{5}\arctan 5x + c$

2) g)

Let $100x^2 = 9\sinh^2 u \Rightarrow 10x = 3\sinh u \Rightarrow x = \frac{3}{10}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{10}\cosh u \Rightarrow \mathrm{d}x = \frac{3}{10}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{100x^2+9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9\sinh^2 u+9}} \times \frac{3}{10}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(\sinh^2 u+1)}} \times \frac{3}{10}\cosh u\,\mathrm{d}u$

$= \frac{1}{10}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{10}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{1}{10}\mathrm{d}u$

$= \frac{1}{10}u + c$

$= \frac{1}{10}\operatorname{arsinh}\left(\frac{10}{3}x\right) + c$

So $x = \frac{3}{10}\sinh u$ gives $\int \frac{1}{\sqrt{100x^2+9}}\mathrm{d}x = \frac{1}{10}\operatorname{arsinh}\left(\frac{10}{3}x\right) + c$

2) h)

Let $121x^2 = 4\sin^2 u \Rightarrow 11x = 2\sin u \Rightarrow x = \frac{2}{11}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2}{11}\cos u \Rightarrow \mathrm{d}x = \frac{2}{11}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{4-121x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{4-4\sin^2 u}} \times \frac{2}{11}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{4(1-\sin^2 u)}} \times \frac{2}{11}\cos u\,\mathrm{d}u$

$= \frac{1}{11}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{11}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{11}\mathrm{d}u$

$= \frac{1}{11}u + c$

$= \frac{1}{11}\arcsin\left(\frac{11}{2}x\right) + c$

So $x = \frac{2}{11}\sin u$ gives $\int \frac{1}{\sqrt{4-121x^2}}\mathrm{d}x = \frac{1}{11}\arcsin\left(\frac{11}{2}x\right) + c$

2) i)

Let $x^2 = 4\cosh^2 u \Rightarrow x = 2\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\sinh u \Rightarrow \mathrm{d}x = 2\sinh u\,\mathrm{d}u$

$\therefore \int \frac{16}{\sqrt{x^2-4}}\mathrm{d}x$

$= \int \frac{16}{\sqrt{4\cosh^2 u-4}} \times 2\sinh u\,\mathrm{d}u$

$= \int \frac{16}{\sqrt{4(\cosh^2 u-1)}} \times 2\sinh u\,\mathrm{d}u$

$= 16\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= 16\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int 16\,\mathrm{d}u$

$= 16u + c$

$= 16\,\mathrm{arcosh}\left(\frac{1}{2}x\right) + c$

So $x = 2\cosh u$ gives $\int \frac{16}{\sqrt{x^2-4}}\mathrm{d}x = 16\,\mathrm{arcosh}\left(\frac{1}{2}x\right) + c$

2) j)

Let $49x^2 = 144\tan^2 u \Rightarrow 7x = 12\tan u \Rightarrow x = \frac{12}{7}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{12}{7}\sec^2 u \Rightarrow \mathrm{d}x = \frac{12}{7}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{7}{144+49x^2}\mathrm{d}x$

$= \int \frac{7}{144+144\tan^2 u} \times \frac{12}{7}\sec^2 u\,\mathrm{d}u$

$= \int \frac{7}{144(1+\tan^2 u)} \times \frac{12}{7}\sec^2 u\,\mathrm{d}u$

$= \frac{1}{12}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{12}\mathrm{d}u$

$= \frac{1}{12}u + c$

$= \frac{1}{12}\arctan\left(\frac{7}{12}x\right) + c$

So $x = \frac{12}{7}\tan u$ gives $\int \frac{7}{144+49x^2}\mathrm{d}x = \frac{1}{12}\arctan\left(\frac{7}{12}x\right) + c$

2) k)

Let $9x^2 = 25\tanh^2 u \Rightarrow 3x = 5\tanh u \Rightarrow x = \frac{5}{3}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5}{3}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{5}{3}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{24}{25-9x^2}\mathrm{d}x$

$= \int \frac{24}{25-25\tanh^2 u} \times \frac{5}{3}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{24}{25(1-\tanh^2 u)} \times \frac{5}{3}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{8}{5}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{8}{5}\mathrm{d}u$

$= \frac{8}{5}u + c$

$= \frac{8}{5}\operatorname{artanh}\left(\frac{3}{5}x\right) + c$

So $x = \frac{5}{3}\tanh u$ gives $\int \frac{24}{25-9x^2}\mathrm{d}x = \frac{8}{5}\operatorname{artanh}\left(\frac{3}{5}x\right) + c$

2) l)

Let $2x^2 = 81\sin^2 u \Rightarrow \sqrt{2}x = 9\sin u \Rightarrow x = \frac{9}{\sqrt{2}}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{9\sqrt{2}}{2}\cos u \Rightarrow \mathrm{d}x = \frac{9\sqrt{2}}{2}\cos u\,\mathrm{d}u$

$\therefore \int \frac{32}{\sqrt{81-2x^2}}\mathrm{d}x$

$= \int \frac{32}{\sqrt{81-81\sin^2 u}} \times \frac{9\sqrt{2}}{2}\cos u\,\mathrm{d}u$

$= \int \frac{32}{\sqrt{81(1-\sin^2 u)}} \times \frac{9\sqrt{2}}{2}\cos u\,\mathrm{d}u$

$= 16\sqrt{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= 16\sqrt{2}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int 16\sqrt{2}\,\mathrm{d}u$

$= 16\sqrt{2}u + c$

$= 16\sqrt{2}\arcsin\left(\frac{\sqrt{2}}{9}x\right) + c$

So $x = \frac{9}{\sqrt{2}}\sin u$ gives $\int \frac{32}{\sqrt{81-2x^2}}\mathrm{d}x = 16\sqrt{2}\arcsin\left(\frac{\sqrt{2}}{9}x\right) + c$

2) m)

Let $9x^2 = 10\cosh^2 u \Rightarrow 3x = \sqrt{10}\cosh u \Rightarrow x = \frac{\sqrt{10}}{3}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{10}}{3}\sinh u \Rightarrow \mathrm{d}x = \frac{\sqrt{10}}{3}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{90}{\sqrt{9x^2-10}}\mathrm{d}x$

$= \int \frac{90}{\sqrt{10\cosh^2 u-10}} \times \frac{\sqrt{10}}{3}\sinh u\,\mathrm{d}u$

$= \int \frac{90}{\sqrt{10(\cosh^2 u-1)}} \times \frac{\sqrt{10}}{3}\sinh u\,\mathrm{d}u$

$= 30\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= 30\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int 30\,\mathrm{d}u$

$= 30u + c$

$= 30\,\mathrm{arcosh}\left(\frac{3}{\sqrt{10}}x\right) + c$

$= 30\,\mathrm{arcosh}\left(\frac{3\sqrt{10}}{10}x\right) + c$

So $x = \frac{\sqrt{10}}{3}\cosh u$ gives $\int \frac{90}{\sqrt{9x^2-10}}\mathrm{d}x = 30\,\mathrm{arcosh}\left(\frac{3\sqrt{10}}{10}x\right) + c$

2) n)

Let $121x^2 = 50\tanh^2 u \Rightarrow 11x = \sqrt{50}\tanh u \Rightarrow x = \frac{\sqrt{50}}{11}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5\sqrt{2}}{11}\mathrm{sech}^2 u \Rightarrow \mathrm{d}x = \frac{5\sqrt{2}}{11}\mathrm{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{55}{50-121x^2}\mathrm{d}x$

$= \int \frac{55}{50-50\tanh^2 u} \times \frac{5\sqrt{2}}{11}\mathrm{sech}^2 u\,\mathrm{d}u$

$= \int \frac{55}{50(1-\tanh^2 u)} \times \frac{5\sqrt{2}}{11}\mathrm{sech}^2 u\,\mathrm{d}u$

$= \frac{\sqrt{2}}{2}\int \frac{\mathrm{sech}^2 u}{\mathrm{sech}^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{2}}{2}\mathrm{d}u$

$= \frac{\sqrt{2}}{2}u + c$

$= \frac{\sqrt{2}}{2}\,\mathrm{artanh}\left(\frac{11}{\sqrt{50}}x\right) + c$

$= \frac{\sqrt{2}}{2}\,\mathrm{artanh}\left(\frac{11\sqrt{2}}{10}x\right) + c$

So $x = \frac{\sqrt{50}}{11}\tanh u$ gives $\int \frac{55}{50-121x^2}\mathrm{d}x = \frac{\sqrt{2}}{2}\,\mathrm{artanh}\left(\frac{11\sqrt{2}}{10}x\right) + c$

2) o)

Let $108x^2 = 5\tan^2 u \Rightarrow 6\sqrt{3}x = \sqrt{5}\tan u \Rightarrow x = \frac{\sqrt{5}}{6\sqrt{3}}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{15}}{18}\sec^2 u \Rightarrow \mathrm{d}x = \frac{\sqrt{15}}{18}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{36}{5+108x^2}\mathrm{d}x$

$= \int \frac{36}{5+5\tan^2 u} \times \frac{\sqrt{15}}{18}\sec^2 u\,\mathrm{d}u$

$= \int \frac{36}{5(1+\tan^2 u)} \times \frac{\sqrt{15}}{18}\sec^2 u\,\mathrm{d}u$

$= \frac{2\sqrt{15}}{5}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{2\sqrt{15}}{5}\mathrm{d}u$

$= \frac{2\sqrt{15}}{5}u + c$

$= \frac{2\sqrt{15}}{5}\arctan\left(\frac{6\sqrt{3}}{\sqrt{5}}x\right) + c$

$= \frac{2\sqrt{15}}{5}\arctan\left(\frac{6\sqrt{15}}{5}x\right) + c$

So $x = \frac{\sqrt{5}}{6\sqrt{3}}\tan u$ gives $\int \frac{36}{5+108x^2}\mathrm{d}x = \frac{2\sqrt{15}}{5}\arctan\left(\frac{6\sqrt{15}}{5}x\right) + c$

3) a)

Let $9x^2 = 4\sinh^2 u \Rightarrow 3x = 2\sinh u \Rightarrow x = \frac{2}{3}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2}{3}\cosh u \Rightarrow \mathrm{d}x = \frac{2}{3}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{3}{\sqrt{9x^2+4}}\mathrm{d}x$

$= \int \frac{3}{\sqrt{4\sinh^2 u+4}} \times \frac{2}{3}\cosh u\,\mathrm{d}u$

$= \int \frac{3}{\sqrt{4(\sinh^2 u+1)}} \times \frac{2}{3}\cosh u\,\mathrm{d}u$

$= \int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \operatorname{arsinh}\left(\frac{3}{2}x\right) + c$

So $x = \frac{2}{3}\sinh u$ gives $\int \frac{3}{\sqrt{9x^2+4}}\mathrm{d}x = \operatorname{arsinh}\left(\frac{3}{2}x\right) + c$

3) b)

Let $49x^2 = 4\tan^2 u \Rightarrow 7x = 2\tan u \Rightarrow x = \frac{2}{7}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2}{7}\sec^2 u \Rightarrow \mathrm{d}x = \frac{2}{7}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{14}{4+49x^2}\mathrm{d}x$

$= \int \frac{14}{4+4\tan^2 u} \times \frac{2}{7}\sec^2 u\,\mathrm{d}u$

$= \int \frac{14}{4(1+\tan^2 u)} \times \frac{2}{7}\sec^2 u\,\mathrm{d}u$

$= \int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \arctan\left(\frac{7}{2}x\right) + c$

So $x = \frac{2}{7}\tan u$ gives $\int \frac{14}{4+49x^2}\mathrm{d}x = \arctan\left(\frac{7}{2}x\right) + c$

3) c)

Let $16x^2 = 9\tanh^2 u \Rightarrow 4x = 3\tanh u \Rightarrow x = \frac{3}{4}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{4}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{3}{4}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{144}{9-16x^2}\mathrm{d}x$

$= \int \frac{144}{9-9\tanh^2 u} \times \frac{3}{4}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{144}{9(1-\tanh^2 u)} \times \frac{3}{4}\operatorname{sech}^2 u\,\mathrm{d}u$

$= 12\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int 12\,\mathrm{d}u$

$= 12u + c$

$= 12\operatorname{artanh}\left(\frac{4}{3}x\right) + c$

So $x = \frac{3}{4}\tanh u$ gives $\int \frac{144}{9-16x^2}\mathrm{d}x = 12\operatorname{artanh}\left(\frac{4}{3}x\right) + c$

3) d)

Let $16x^2 = 9\cosh^2 u \Rightarrow 4x = 3\cosh u \Rightarrow x = \frac{3}{4}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{4}\sinh u \Rightarrow \mathrm{d}x = \frac{3}{4}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{8}{\sqrt{16x^2-9}}\mathrm{d}x$

$= \int \frac{8}{\sqrt{9\cosh^2 u-9}} \times \frac{3}{4}\sinh u\,\mathrm{d}u$

$= \int \frac{8}{\sqrt{9(\cosh^2 u-1)}} \times \frac{3}{4}\sinh u\,\mathrm{d}u$

$= 2\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= 2\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int 2\,\mathrm{d}u$

$= 2u + c$

$= 2\operatorname{arcosh}\left(\frac{4}{3}x\right) + c$

So $x = \frac{3}{4}\cosh u$ gives $\int \frac{8}{\sqrt{16x^2-9}}\mathrm{d}x = 2\operatorname{arcosh}\left(\frac{4}{3}x\right) + c$

3) e)

Let $4x^2 = 25\sin^2 u \Rightarrow 2x = 5\sin u \Rightarrow x = \frac{5}{2}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5}{2}\cos u \Rightarrow \mathrm{d}x = \frac{5}{2}\cos u\,\mathrm{d}u$

$\therefore I = \int \sqrt{25-4x^2}\,\mathrm{d}x$

$= \int \sqrt{25-25\sin^2 u} \times \frac{5}{2}\cos u\,\mathrm{d}u$

$= \int \sqrt{25(1-\sin^2 u)} \times \frac{5}{2}\cos u\,\mathrm{d}u$

$= \frac{25}{2}\int \sqrt{1-\sin^2 u} \times \cos u\,\mathrm{d}u$

$= \frac{25}{2}\int \sqrt{\cos^2 u} \times \cos u\,\mathrm{d}u$

$= \frac{25}{2}\int \cos^2 u\,\mathrm{d}u$

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore I = \frac{25}{2}\int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int \left(\frac{25}{4}\cos 2u + \frac{25}{4}\right)\mathrm{d}u$

$= \frac{25}{8}\sin 2u + \frac{25}{4}u + c$

$= \frac{25}{8}(2\sin u\cos u) + \frac{25}{4}u + c$

$= \frac{25}{4}\sin u \cos u + \frac{25}{4}u + c$

$= \frac{25}{4}\sin u \sqrt{1-\sin^2 u} + \frac{25}{4}u + c$

$= \frac{25}{4}\left(\frac{2}{5}x\right)\sqrt{1-\left(\frac{2}{5}x\right)^2} + \frac{25}{4}\arcsin\left(\frac{2}{5}x\right) + c$

$= \frac{5}{2}x\sqrt{1-\frac{4}{25}x^2} + \frac{25}{4}\arcsin\left(\frac{2}{5}x\right) + c$

$= \frac{5}{2}x\sqrt{\frac{1}{25}(25-4x^2)} + \frac{25}{4}\arcsin\left(\frac{2}{5}x\right) + c$

$= \frac{1}{2}x\sqrt{25-4x^2} + \frac{25}{4}\arcsin\left(\frac{2}{5}x\right) + c$

So $x = \frac{5}{2}\sin u$ gives $\int \sqrt{25-4x^2}\,\mathrm{d}x = \frac{1}{2}x\sqrt{25-4x^2} + \frac{25}{4}\arcsin\left(\frac{2}{5}x\right) + c$

3) f)

Let $9x^2 = 16\tanh^2 u \Rightarrow 3x = 4\tanh u \Rightarrow x = \frac{4}{3}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{4}{3}\mathrm{sech}^2 u \Rightarrow \mathrm{d}x = \frac{4}{3}\mathrm{sech}^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{24576}{(16-9x^2)^3}\,\mathrm{d}x$

$= \int \frac{24576}{(16-16\tanh^2 u)^3} \times \frac{4}{3}\mathrm{sech}^2 u\,\mathrm{d}u$

$= \int \frac{24576}{\left(16(1-\tanh^2 u)\right)^3} \times \frac{4}{3}\mathrm{sech}^2 u\,\mathrm{d}u$

$= 8\int \frac{\mathrm{sech}^2 u}{(\mathrm{sech}^2 u)^3}\,\mathrm{d}u$

$= 8\int \frac{1}{(\mathrm{sech}^2 u)^2}\,\mathrm{d}u$

$= 8\int \frac{1}{\mathrm{sech}^4 u}\,\mathrm{d}u$

$= 8\int \cosh^4 u\,\mathrm{d}u$

Now:

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore \cosh^4 u$

$= (\cosh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = 8\int\left(\frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int(\cosh 4u + 4\cosh 2u + 3)\,\mathrm{d}u$

$= \frac{1}{4}\sinh 4u + 2\sinh 2u + 3u + c$

$= \frac{1}{4}(2\sinh 2u\cosh 2u) + 2(2\sinh u\cosh u) + 3u + c$

$= \frac{1}{2}\sinh 2u\cosh 2u + 4\sinh u\cosh u + 3u + c$

$= \frac{1}{2}(2\sinh u\cosh u)(2\cosh^2 u - 1) + 4\sinh u\cosh u + 3u + c$

$= 2\sinh u\cosh^3 u - \sinh u\cosh u + 4\sinh u\cosh u + 3u + c$

$= 2\sinh u\cosh^3 u + 3\sinh u\cosh u + 3u + c$

If $\tanh u = \frac{3}{4}x$

$\Rightarrow \frac{\sinh u}{\cosh u} = \frac{3}{4}x$

$\Rightarrow \sinh u = \frac{3}{4}x\cosh u$

Use the identity $\cosh^2 A - \sinh^2 A \equiv 1$

$\therefore \cosh^2 u - \left(\frac{3}{4}x\cosh u\right)^2 = 1$

$\Rightarrow \cosh^2 u - \frac{9}{16}x\cosh^2 u = 1$

$\Rightarrow \cosh^2 u\left(1 - \frac{9}{16}x^2\right) = 1$

$\Rightarrow \cosh^2 u = \frac{1}{1-\frac{9}{16}x^2} = \frac{16}{16-9x^2}$

$\Rightarrow \cosh u = \frac{4}{\sqrt{16-9x^2}}$ as $\cosh u > 0$

$\Rightarrow \sinh u = \frac{3x}{\sqrt{16-9x^2}}$

$\therefore I = 2\times\frac{3x}{\sqrt{16-9x^2}}\times\left(\frac{4}{\sqrt{16-9x^2}}\right)^3 + 3\times\frac{3x}{\sqrt{16-9x^2}}\times\frac{4}{\sqrt{16-9x^2}} + 3\,\mathrm{artanh}\left(\frac{3}{4}x\right) + c$

$= \frac{384x}{(16-9x^2)^2} + \frac{36x}{(16-9x^2)} + 3\,\mathrm{artanh}\left(\frac{3}{4}x\right) + c$

$= \frac{384x}{(16-9x^2)^2} + \frac{36x(16-9x^2)}{(16-9x^2)^2} + 3\,\mathrm{artanh}\left(\frac{3}{4}x\right) + c$

$= \frac{384x+36x(16-9x^2)}{(16-9x^2)^2} + 3\,\mathrm{artanh}\left(\frac{3}{4}x\right) + c$

$= \frac{960x-324x^3}{(16-9x^2)^2} + 3\,\mathrm{artanh}\left(\frac{3}{4}x\right) + c$

So $x = \frac{4}{3}\tanh u$ gives $\int\frac{24576}{(16-9x^2)^3}\mathrm{d}x = \frac{960x-324x^3}{(16-9x^2)^2} + 3\,\mathrm{artanh}\left(\frac{3}{4}x\right) + c$

3) g)

Let $49x^2 = 4\sin^2 u \Rightarrow 7x = 2\sin u \Rightarrow x = \frac{2}{7}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2}{7}\cos u \Rightarrow \mathrm{d}x = \frac{2}{7}\cos u\,\mathrm{d}u$

$\therefore I = \int (4 - 49x^2)^{\frac{3}{2}}\,\mathrm{d}x$

$= \int (4 - 4\sin^2 u)^{\frac{3}{2}} \times \frac{2}{7}\cos u\,\mathrm{d}u$

$= \int \left(4(1 - \sin^2 u)\right)^{\frac{3}{2}} \times \frac{2}{7}\cos u\,\mathrm{d}u$

$= \frac{16}{7}\int (\cos^2 u)^{\frac{3}{2}} \times \cos u\,\mathrm{d}u$

$= \frac{16}{7}\int \cos^4 u\,\mathrm{d}u$

Now:

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A = \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore \cos^4 u$

$= (\cos^2 u)^2$

$= \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cos^2 2u + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cos 4u + \frac{1}{2}\right) + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{8} + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}$

$\therefore I = \frac{16}{7}\int \left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int \left(\frac{2}{7}\cos 4u + \frac{8}{7}\cos 2u + \frac{6}{7}\right)\mathrm{d}u$

$= \frac{1}{14}\sin 4u + \frac{4}{7}\sin 2u + \frac{6}{7}u + c$

$= \frac{1}{14}(2\sin 2u\cos 2u) + \frac{4}{7}(2\sin u\cos u) + \frac{6}{7}u + c$

$= \frac{1}{7}\sin 2u\cos 2u + \frac{8}{7}\sin u\cos u + \frac{6}{7}u + c$

$= \frac{1}{7}(2\sin u\cos u)(2\cos^2 u - 1) + \frac{8}{7}\sin u\cos u + \frac{6}{7}u + c$

$= \frac{4}{7}\sin u\cos^3 u - \frac{2}{7}\sin u\cos u + \frac{8}{7}\sin u\cos u + \frac{6}{7}u + c$

$= \frac{4}{7}\sin u\cos^3 u + \frac{6}{7}\sin u\cos u + \frac{6}{7}u + c$

If $\sin u = \frac{7x}{2}$ then $\cos u = \frac{\sqrt{4-49x^2}}{2}$

$\therefore I = \frac{4}{7} \times \frac{7x}{2} \times \left(\frac{\sqrt{4-49x^2}}{2}\right)^3 + \frac{6}{7} \times \frac{7x}{2} \times \frac{\sqrt{4-49x^2}}{2} + \frac{6}{7}\arcsin\left(\frac{7}{2}x\right) + c$

$= \frac{1}{4}x(4-49x^2)^{\frac{3}{2}} + \frac{3}{2}x\sqrt{4-49x^2} + \frac{6}{7}\arcsin\left(\frac{7}{2}x\right) + c$

$= \frac{1}{4}x\big((4-49x^2)+6\big)\sqrt{4-49x^2} + \frac{6}{7}\arcsin\left(\frac{7}{2}x\right) + c$

$= \frac{1}{4}x(10-49x^2)\sqrt{4-49x^2} + \frac{6}{7}\arcsin\left(\frac{7}{2}x\right) + c$

So $x = \frac{2}{7}\sin u$ gives $\int \frac{14}{4+49x^2}\mathrm{d}x = \frac{1}{4}x(10-49x^2)\sqrt{4-49x^2} + \frac{6}{7}\arcsin\left(\frac{7}{2}x\right) + c$

3) h)

Let $36x^2 = 25\sinh^2 u \Rightarrow 6x = 5\sinh u \Rightarrow x = \frac{5}{6}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5}{6}\cosh u \Rightarrow \mathrm{d}x = \frac{5}{6}\cosh u\,\mathrm{d}u$

$\therefore I = \int \sqrt{36x^2 + 25}\,\mathrm{d}x$

$= \int \sqrt{25\sinh^2 u + 25} \times \frac{5}{6}\cosh u\,\mathrm{d}u$

$= \int \sqrt{25(\sinh^2 u + 1)} \times \frac{5}{6}\cosh u\,\mathrm{d}u$

$= \frac{25}{6}\int \sqrt{\cosh^2 u} \times \cosh u\,\mathrm{d}u$

$= \frac{25}{6}\int \cosh^2 u\,\mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A \equiv \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore I = \frac{25}{6}\int \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int \left(\frac{25}{12}\cosh 2u + \frac{25}{12}\right)\mathrm{d}u$

$= \frac{25}{24}\sinh 2u + \frac{25}{12}u + c$

$= \frac{25}{24}(2\sinh u\cosh u) + \frac{25}{12}u + c$

$= \frac{25}{12}\sinh u\cosh u + \frac{25}{12}u + c$

$= \frac{25}{12}\sinh u\sqrt{\sinh^2 u + 1} + \frac{25}{12}u + c$

$= \frac{25}{12}\left(\frac{6}{5}x\right)\sqrt{\left(\frac{6}{5}x\right)^2 + 1} + \frac{25}{12}\operatorname{arsinh}\left(\frac{6}{5}x\right) + c$

$= \frac{5}{2}x\sqrt{\frac{36}{25}x^2 + 1} + \frac{25}{12}\operatorname{arsinh}\left(\frac{6}{5}x\right) + c$

$= \frac{5}{2}x\sqrt{\frac{1}{25}(36x^2 + 25)} + \frac{25}{12}\operatorname{arsinh}\left(\frac{6}{5}x\right) + c$

$= \frac{1}{2}x\sqrt{36x^2 + 25} + \frac{25}{12}\operatorname{arsinh}\left(\frac{6}{5}x\right) + c$

So $x = \frac{5}{6}\sinh u$ gives $\int \sqrt{36x^2 + 25}\,\mathrm{d}x = \frac{1}{2}x\sqrt{36x^2 + 25} + \frac{25}{12}\operatorname{arsinh}\left(\frac{6}{5}x\right) + c$

3) i)

Let $9x^2 = 100\tan^2 u \Rightarrow 3x = 10\tan u \Rightarrow x = \frac{10}{3}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{10}{3}\sec^2 u \Rightarrow \mathrm{d}x = \frac{10}{3}\sec^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{900000}{(100+9x^2)^3}\mathrm{d}x$

$= \int \frac{900000}{(100+100\tan^2 u)^3} \times \frac{10}{3}\sec^2 u\,\mathrm{d}u$

$= \int \frac{900000}{\left(100(1+\tan^2 u)\right)^3} \times \frac{10}{3}\sec^2 u\,\mathrm{d}u$

$= 3\int \frac{\sec^2 u}{(\sec^2 u)^3}\mathrm{d}u$

$= 3\int \frac{1}{(\sec^2 u)^2}\mathrm{d}u$

$= 3\int \frac{1}{\sec^4 u}\mathrm{d}u$

$= 3\int \cos^4 u\,\mathrm{d}u$

Now:

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A = \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore \cos^4 u$

$= (\cos^2 u)^2$

$= \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cos^2 2u + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cos 4u + \frac{1}{2}\right) + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{8} + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}$

$\therefore I = 3\int \left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int \left(\frac{3}{8}\cos 4u + \frac{3}{2}\cos 2u + \frac{9}{8}\right)\mathrm{d}u$

$= \frac{3}{32}\sin 4u + \frac{3}{4}\sin 2u + \frac{9}{8}u + c$

$= \frac{3}{32}(2\sin 2u\cos 2u) + \frac{3}{4}(2\sin u\cos u) + \frac{9}{8}u + c$

$= \frac{3}{16}\sin 2u\cos 2u + \frac{3}{2}\sin u\cos u + \frac{9}{8}u + c$

$= \frac{3}{16}(2\sin u\cos u)(2\cos^2 u - 1) + \frac{3}{2}\sin u\cos u + \frac{9}{8}u + c$

$= \frac{3}{4}\sin u\cos^3 u - \frac{3}{8}\sin u\cos u + \frac{3}{2}\sin u\cos u + \frac{9}{8}u + c$

$= \frac{3}{4}\sin u\cos^3 u + \frac{9}{8}\sin u\cos u + \frac{9}{8}u + c$

If $\tan u = \frac{3x}{10}$ then $\sin u = \frac{3x}{\sqrt{100+9x^2}}$ and $\cos u = \frac{10}{\sqrt{100+9x^2}}$

$\therefore I = \frac{3}{4} \times \frac{3x}{\sqrt{100+9x^2}} \times \left(\frac{10}{\sqrt{100+9x^2}}\right)^3 + \frac{9}{8} \times \frac{3x}{\sqrt{100+9x^2}} \times \frac{10}{\sqrt{100+9x^2}} + \frac{9}{8}\arctan\left(\frac{3}{10}x\right) + c$

$= \frac{2250x}{(100+9x^2)^2} + \frac{135x}{4(100+9x^2)} + \frac{9}{8}\arctan\left(\frac{3}{10}x\right) + c$

$= \frac{9000x}{4(100+9x^2)^2} + \frac{135x(100+9x^2)}{4(100+9x^2)^2} + \frac{9}{8}\arctan\left(\frac{3}{10}x\right) + c$

$= \frac{9000x+135x(100+9x^2)}{4(100+9x^2)^2} + \frac{9}{8}\arctan\left(\frac{3}{10}x\right) + c$

$= \frac{22500x+1215x^3}{4(100+9x^2)^2} + \frac{9}{8}\arctan\left(\frac{3}{10}x\right) + c$

So $x = \frac{10}{3}\tan u$ gives $\int \frac{900000}{(100+9x^2)^3}\,\mathrm{d}x = \frac{22500x+1215x^3}{4(100+9x^2)^2} + \frac{9}{8}\arctan\left(\frac{3}{10}x\right) + c$

3) j)

Let $49x^2 = 25\tanh^2 u \Rightarrow 7x = 5\tanh u \Rightarrow x = \frac{5}{7}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5}{7}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{5}{7}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{280}{(25-49x^2)^2}\,\mathrm{d}x$

$= \int \frac{280}{(25-25\tanh^2 u)^2} \times \frac{5}{7}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{280}{\left(25(1-\tanh^2 u)\right)^2} \times \frac{5}{7}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{8}{25}\int \frac{\operatorname{sech}^2 u}{(\operatorname{sech}^2 u)^2}\,\mathrm{d}u$

$= \frac{8}{25}\int \frac{1}{\operatorname{sech}^2 u}\,\mathrm{d}u$

$= \frac{8}{25}\int \cosh^2 u\,\mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A \equiv \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore I = \frac{8}{25}\int \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int \left(\frac{4}{25}\cosh 2u + \frac{4}{25}\right)\mathrm{d}u$

$= \frac{2}{25}\sinh 2u + \frac{4}{25}u + c$

$= \frac{2}{25}(2\sinh u \cosh u) + \frac{4}{25}u + c$

$= \frac{4}{25}\sinh u \cosh u + \frac{4}{25}u + c$

If $\tanh u = \frac{7}{5}x$

$\Rightarrow \frac{\sinh u}{\cosh u} = \frac{7}{5}x$

$\Rightarrow \sinh u = \frac{7}{5}x\cosh u$

Use the identity $\cosh^2 A - \sinh^2 A \equiv 1$

$\therefore \cosh^2 u - \left(\frac{7}{5}x\cosh u\right)^2 = 1$

$\Rightarrow \cosh^2 u - \frac{49}{25}x^2\cosh^2 u = 1$

$\Rightarrow \cosh^2 u\left(1 - \frac{49}{25}x^2\right) = 1$

$\Rightarrow \cosh^2 u = \frac{1}{1-\frac{49}{25}x^2} = \frac{25}{25-49x^2}$

$\Rightarrow \cosh u = \frac{5}{\sqrt{25-49x^2}}$ as $\cosh u > 0$

$\Rightarrow \sinh u = \frac{7x}{\sqrt{25-49x^2}}$

$\therefore I = \frac{4}{25} \times \frac{7x}{\sqrt{25-49x^2}} \times \frac{5}{\sqrt{25-49x^2}} + \frac{4}{25}\operatorname{artanh}\left(\frac{7}{5}x\right) + c$

$= \frac{28x}{5(25-49x^2)} + \frac{4}{25}\operatorname{artanh}\left(\frac{7}{5}x\right) + c$

So $x = \frac{5}{7}\tanh u$ gives $\int \frac{280}{(25-49x^2)^2}\,\mathrm{d}x = \frac{28x}{5(25-49x^2)} + \frac{4}{25}\operatorname{artanh}\left(\frac{7}{5}x\right) + c$

3) k)

Let $2x^2 = 25\tan^2 u \Rightarrow \sqrt{2}x = 5\tan u \Rightarrow x = \frac{5}{\sqrt{2}}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5\sqrt{2}}{2}\sec^2 u \Rightarrow \mathrm{d}x = \frac{5\sqrt{2}}{2}\sec^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{250}{(25+2x^2)^2}\,\mathrm{d}x$

$= \int \frac{250}{(25+25\tan^2 u)^2} \times \frac{5\sqrt{2}}{2}\sec^2 u\,\mathrm{d}u$

$= \int \frac{250}{\left(25(1+\tan^2 u)\right)^2} \times \frac{5\sqrt{2}}{2}\sec^2 u\,\mathrm{d}u$

$= \sqrt{2}\int \frac{\sec^2 u}{(\sec^2 u)^2}\,\mathrm{d}u$

$= \sqrt{2}\int \frac{1}{\sec^2 u}\,\mathrm{d}u$

$= \sqrt{2}\int \cos^2 u\,\mathrm{d}u$

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore I = \sqrt{2}\int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int \left(\frac{\sqrt{2}}{2}\cos 2u + \frac{\sqrt{2}}{2}\right)\mathrm{d}u$

$= \frac{\sqrt{2}}{4}\sin 2u + \frac{\sqrt{2}}{2}u + c$

$= \frac{\sqrt{2}}{4}(2\sin u\cos u) + \frac{\sqrt{2}}{2}u + c$

$= \frac{\sqrt{2}}{2}\sin u\cos u + \frac{\sqrt{2}}{2}u + c$

If $\tan u = \frac{\sqrt{2}x}{5}$ then $\sin u = \frac{\sqrt{2}x}{\sqrt{25+2x^2}}$ and $\cos u = \frac{5}{\sqrt{25+2x^2}}$

$\therefore I = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}x}{\sqrt{25+2x^2}} \times \frac{5}{\sqrt{25+2x^2}} + \frac{\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{5}x\right) + c$

$= \frac{5x}{25+2x^2} + \frac{\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{5}x\right) + c$

So $x = \frac{5}{\sqrt{2}}\tan u$ gives $\int \frac{250}{(25+2x^2)^2}\,\mathrm{d}x = \frac{5x}{25+2x^2} + \frac{\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{5}x\right) + c$

3) l)

Let $3x^2 = 4\cosh^2 u \Rightarrow \sqrt{3}x = 2\cosh u \Rightarrow x = \frac{2}{\sqrt{3}}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{3}}{3}\sinh u \Rightarrow \mathrm{d}x = \frac{2\sqrt{3}}{3}\sinh u\,\mathrm{d}u$

$\therefore I = \int (3x^2 - 4)^{\frac{3}{2}}\,\mathrm{d}x$

$= \int (4\cosh^2 u - 4)^{\frac{3}{2}} \times \frac{2\sqrt{3}}{3}\sinh u\,\mathrm{d}u$

$= \int \left(4(\cosh^2 u - 1)\right)^{\frac{3}{2}} \times \frac{2\sqrt{3}}{3}\sinh u\,\mathrm{d}u$

$= \frac{16\sqrt{3}}{3}\int (\sinh^2 u)^{\frac{3}{2}} \times \sinh u\,\mathrm{d}u$

$= \frac{16\sqrt{3}}{3}\int \sinh^3 u \times \sinh u\,\mathrm{d}u$

$= \frac{16\sqrt{3}}{3}\int \sinh^4 u\,\mathrm{d}u$

Now:

$\cosh 2A \equiv 1 + 2\sinh^2 A \Leftrightarrow \sinh^2 A = \frac{1}{2}\cosh 2A - \frac{1}{2}$

$\therefore \sinh^4 u$

$= (\sinh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u - \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = \frac{16\sqrt{3}}{3}\int \left(\frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int \left(\frac{2\sqrt{3}}{3}\cosh 4u - \frac{8\sqrt{3}}{3}\cosh 2u + 2\sqrt{3}\right)\mathrm{d}u$

$= \frac{\sqrt{3}}{6}\sinh 4u - \frac{4\sqrt{3}}{3}\sinh 2u + 2\sqrt{3}u + c$

$= \frac{\sqrt{3}}{6}(2\sinh 2u\cosh 2u) - \frac{4\sqrt{3}}{3}(2\sinh u\cosh u) + 2\sqrt{3}u + c$

$= \frac{\sqrt{3}}{3}(2\sinh u\cosh u)(1 + 2\sinh^2 u) - \frac{8\sqrt{3}}{3}\sinh u\cosh u + 2\sqrt{3}u + c$

$= \frac{\sqrt{3}}{3}(2\sinh u\cosh u + 4\sinh^3 u\cosh u) - \frac{8\sqrt{3}}{3}\sinh u\cosh u + 2\sqrt{3}u + c$

$= \frac{2\sqrt{3}}{3}\sinh u\cosh u + \frac{4\sqrt{3}}{3}\sinh^3 u\cosh u - \frac{8\sqrt{3}}{3}\sinh u\cosh u + 2\sqrt{3}u + c$

$= -2\sqrt{3}\sinh u\cosh u + \frac{4\sqrt{3}}{3}\sinh^3 u\cosh u + 2\sqrt{3}u + c$

$= -2\sqrt{3}\sqrt{\cosh^2 u - 1}\cosh u + \frac{4\sqrt{3}}{3}\left(\sqrt{\cosh^2 u - 1}\right)^3\cosh u + 2\sqrt{3}u + c$

$= -2\sqrt{3}\left(\frac{\sqrt{3}}{2}x\right)\sqrt{\left(\frac{\sqrt{3}}{2}x\right)^2 - 1} + \frac{4\sqrt{3}}{3}\left(\frac{\sqrt{3}}{2}x\right)\left(\left(\frac{\sqrt{3}}{2}x\right)^2 - 1\right)^{\frac{3}{2}} + 2\sqrt{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x\right) + c$

$= -3x\sqrt{\frac{3}{4}x^2 - 1} + 2x\left(\frac{3}{4}x^2 - 1\right)^{\frac{3}{2}} + 2\sqrt{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x\right) + c$

$= -3x\sqrt{\frac{1}{4}(3x^2 - 4)} + 2x\left(\frac{1}{4}(3x^2 - 4)\right)^{\frac{3}{2}} + 2\sqrt{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x\right) + c$

$= -\frac{3}{2}x\sqrt{3x^2 - 4} + \frac{1}{4}x(3x^2 - 4)^{\frac{3}{2}} + 2\sqrt{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x\right) + c$

$= \frac{1}{4}x\big(-6 + (3x^2 - 4)\big)\sqrt{3x^2 - 4} + 2\sqrt{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x\right) + c$

$= \frac{1}{4}x(3x^2 - 10)\sqrt{3x^2 - 4} + 2\sqrt{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x\right) + c$

So $x = \frac{2}{\sqrt{3}}\cosh u$

gives $\int (3x^2 - 4)^{\frac{3}{2}}\,\mathrm{d}x = \frac{1}{4}x(3x^2 - 10)\sqrt{3x^2 - 4} + 2\sqrt{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x\right) + c$

4) a)

Let $(3x-1)^2 = \sin^2 u \Rightarrow 3x-1 = \sin u \Rightarrow x = 1 + \frac{1}{3}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{3}\cos u \Rightarrow \mathrm{d}x = \frac{1}{3}\cos u \,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{1-(3x-1)^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{1-\sin^2 u}} \times \frac{1}{3}\cos u \,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{3}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{3}\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\arcsin(3x-1) + c$

So $x = 1 + \frac{1}{3}\sin u$ gives $\int \frac{1}{\sqrt{1-(3x-1)^2}}\mathrm{d}x = \frac{1}{3}\arcsin(3x-1) + c$

4) b)

Let $(x-1)^2 = 9\sinh^2 u \Rightarrow x-1 = 3\sinh u \Rightarrow x = 1 + 3\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\cosh u \Rightarrow \mathrm{d}x = 3\cosh u \,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{(x-1)^2+9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9\sinh^2 u+9}} \times 3\cosh u \,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(\sinh^2 u+1)}} \times 3\cosh u \,\mathrm{d}u$

$= \frac{3}{3}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int 1 \,\mathrm{d}u$

$= u + c$

$= \operatorname{arsinh}\left(\frac{x-1}{3}\right) + c$

So $x = 1 + 3\sinh u$ gives $\int \frac{1}{\sqrt{(x-1)^2+9}}\mathrm{d}x = \operatorname{arsinh}\left(\frac{x-1}{3}\right) + c$

4) c)

Let $(2x+5)^2 = 3\sin^2 u \Rightarrow 2x+5 = \sqrt{3}\sin u \Rightarrow x = -\frac{5}{2} + \frac{\sqrt{3}}{2}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{3}}{2}\cos u \Rightarrow \mathrm{d}x = \frac{\sqrt{3}}{2}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{3-(2x+5)^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{3-3\sin^2 u}} \times \frac{\sqrt{3}}{2}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{3(1-\sin^2 u)}} \times \frac{\sqrt{3}}{2}\cos u\,\mathrm{d}u$

$= \frac{\sqrt{3}}{2\sqrt{3}}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{2}\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\arcsin\left(\frac{2x+5}{\sqrt{3}}\right) + c$

So $x = -\frac{5}{2} + \frac{\sqrt{3}}{2}\sin u$ gives $\int \frac{1}{\sqrt{3-(2x+5)^2}}\mathrm{d}x = \frac{1}{2}\arcsin\left(\frac{2x+5}{\sqrt{3}}\right) + c$

4) d)

Let $(3x-2)^2 = 8\sinh^2 u \Rightarrow 3x-2 = 2\sqrt{2}\sinh u \Rightarrow x = \frac{2}{3} + \frac{2\sqrt{2}}{3}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{2}}{3}\cosh u \Rightarrow \mathrm{d}x = \frac{2\sqrt{2}}{3}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{(3x-2)^2+8}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{8\sinh^2 u+8}} \times \frac{2\sqrt{2}}{3}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{8(\sinh^2 u+1)}} \times \frac{2\sqrt{2}}{3}\cosh u\,\mathrm{d}u$

$= \frac{2\sqrt{2}}{3\sqrt{8}}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{3}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \frac{1}{3}\int 1\,\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\mathrm{arsinh}\left(\frac{3x-2}{2\sqrt{2}}\right) + c$

So $x = \frac{2}{3} + \frac{2\sqrt{2}}{3}\sinh u$ gives $\int \frac{1}{\sqrt{(3x-2)^2+8}}\mathrm{d}x = \frac{1}{3}\mathrm{arsinh}\left(\frac{3x-2}{2\sqrt{2}}\right) + c$

4) e)

Let $(4x-1)^2 = 8\tan^2 u \Rightarrow 4x-1 = 2\sqrt{2}\tan u \Rightarrow x = \frac{1}{4} + \frac{\sqrt{2}}{2}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{2}}{2}\sec^2 u \Rightarrow \mathrm{d}x = \frac{\sqrt{2}}{2}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{8+(4x-1)^2}\mathrm{d}x$

$= \int \frac{1}{8+8\tan^2 u} \times \frac{\sqrt{2}}{2}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{8(1+\tan^2 u)} \times \frac{\sqrt{2}}{2}\sec^2 u\,\mathrm{d}u$

$= \frac{\sqrt{2}}{16}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{2}}{16}\mathrm{d}u$

$= \frac{\sqrt{2}}{16}u + c$

$= \frac{\sqrt{2}}{16}\arctan\left(\frac{4x-1}{2\sqrt{2}}\right) + c$

So $x = \frac{1}{4} + \frac{\sqrt{2}}{2}\tan u$ gives $\int \frac{1}{8+(4x-1)^2}\mathrm{d}x = \frac{\sqrt{2}}{16}\arctan\left(\frac{4x-1}{2\sqrt{2}}\right) + c$

4) f)

Let $(5x+3)^2 = 11\tanh^2 u \Rightarrow 5x+3 = \sqrt{11}\tanh u \Rightarrow x = -\frac{3}{5} + \frac{\sqrt{11}}{5}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{11}}{5}\mathrm{sech}^2 u \Rightarrow \mathrm{d}x = \frac{\sqrt{11}}{5}\mathrm{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{4}{11-(5x+3)^2}\mathrm{d}x$

$= \int \frac{4}{11-11\tanh^2 u} \times \frac{\sqrt{11}}{5}\mathrm{sech}^2 u\,\mathrm{d}u$

$= \int \frac{4}{11(1-\tanh^2 u)} \times \frac{\sqrt{11}}{5}\mathrm{sech}^2 u\,\mathrm{d}u$

$= \frac{4\sqrt{11}}{55}\int \frac{\mathrm{sech}^2 u}{\mathrm{sech}^2 u}\mathrm{d}u$

$= \int \frac{4\sqrt{11}}{55}\mathrm{d}u$

$= \frac{4\sqrt{11}}{55}u + c$

$= \frac{4\sqrt{11}}{55}\mathrm{artanh}\left(\frac{5x+3}{\sqrt{11}}\right) + c$

So $x = -\frac{3}{5} + \frac{\sqrt{11}}{5}\tanh u$ gives $\int \frac{4}{11-(5x+3)^2}\mathrm{d}x = \frac{4\sqrt{11}}{55}\mathrm{artanh}\left(\frac{5x+3}{\sqrt{11}}\right) + c$

5) a)

First: $4x^2 + 12x + 10 = 4\left(x + \frac{3}{2}\right)^2 + 1$

Let $4\left(x + \frac{3}{2}\right)^2 = \tan^2 u \Rightarrow 2\left(x + \frac{3}{2}\right) = \tan u \Rightarrow x = -\frac{3}{2} + \frac{1}{2}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\sec^2 u \Rightarrow \mathrm{d}x = \frac{1}{2}\sec^2 u \,\mathrm{d}u$

$\therefore \int \frac{1}{4x^2+12x+10}\mathrm{d}x$

$= \int \frac{1}{4\left(x+\frac{3}{2}\right)^2+1}\mathrm{d}x$

$= \int \frac{1}{\tan^2 u+1} \times \frac{1}{2}\sec^2 u \,\mathrm{d}u$

$= \frac{1}{2}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{2}\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\arctan\left(2\left(x + \frac{3}{2}\right)\right) + c$

So $x = -\frac{3}{2} + \frac{1}{2}\tan u$ gives $\int \frac{1}{4x^2+12x+10}\mathrm{d}x = \frac{1}{2}\arctan(2x+3) + c$

5) b)

First: $3 - 4x - 4x^2 = 4 - 4\left(x + \frac{1}{2}\right)^2$

Let $4\left(x + \frac{1}{2}\right)^2 = 4\sin^2 u \Rightarrow 2\left(x + \frac{1}{2}\right) = 2\sin u \Rightarrow x = -\frac{1}{2} + \sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \,\mathrm{d}u$

$\therefore \int \frac{3}{\sqrt{3-4x-4x^2}}\mathrm{d}x$

$= \int \frac{3}{\sqrt{4-4\sin^2 u}} \times \cos u \,\mathrm{d}u$

$= \int \frac{3}{\sqrt{4(1-\sin^2 u)}} \times \cos u \,\mathrm{d}u$

$= \frac{3}{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{3}{2}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{3}{2}\mathrm{d}u$

$= \frac{3}{2}u + c$

$= \frac{3}{2}\arcsin\left(x + \frac{1}{2}\right) + c$

So $x = -\frac{1}{2} + \sin u$ gives $\int \frac{3}{\sqrt{3-4x-4x^2}}\mathrm{d}x = \frac{3}{2}\arcsin\left(x + \frac{1}{2}\right) + c$

5) c)

First: $x^2 - 6x + 14 = (x-3)^2 + 5$

Let $(x-3)^2 = 5\tan^2 u \Rightarrow x - 3 = \sqrt{5}\tan u \Rightarrow x = 3 + \sqrt{5}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sqrt{5}\sec^2 u \Rightarrow \mathrm{d}x = \sqrt{5}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{2}{x^2-6x+14}\mathrm{d}x$

$= \int \frac{2}{(x-3)^2+5}\mathrm{d}x$

$\int \frac{2}{5\tan^2 u+5} \times \sqrt{5}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{5(\tan^2 u+1)} \times \sqrt{5}\sec^2 u\,\mathrm{d}u$

$= \frac{2\sqrt{5}}{5}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{2\sqrt{5}}{5}\mathrm{d}u$

$= \frac{2\sqrt{5}}{5}u + c$

$= \frac{2\sqrt{5}}{5}\arctan\left(\frac{x-3}{\sqrt{5}}\right) + c$

So $x = 3 + \sqrt{5}\tan u$ gives $\int \frac{2}{x^2-6x+14}\mathrm{d}x = \frac{2\sqrt{5}}{5}\arctan\left(\frac{x-3}{\sqrt{5}}\right) + c$

5) d)

First: $7 + 4x - x^2 = 11 - (x-2)^2$

Let $(x-2)^2 = 11\sin^2 u \Rightarrow x - 2 = \sqrt{11}\sin u \Rightarrow x = 2 + \sqrt{11}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sqrt{11}\cos u \Rightarrow \mathrm{d}x = \sqrt{11}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{7+4x-x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{11-(x-2)^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{11-11\sin^2 u}} \times \sqrt{11}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{11(1-\sin^2 u)}} \times \sqrt{11}\cos u\,\mathrm{d}u$

$= \frac{\sqrt{11}}{\sqrt{11}}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{\sqrt{11}}{\sqrt{11}}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \arcsin\left(\frac{x-2}{\sqrt{11}}\right) + c$

So $x = 2 + \sqrt{11}\sin u$ gives $\int \frac{1}{\sqrt{7+4x-x^2}}\mathrm{d}x = \arcsin\left(\frac{x-2}{\sqrt{11}}\right) + c$

5) e)

First: $3x^2 + 8x + 6 = 3\left(x + \frac{4}{3}\right)^2 + \frac{2}{3}$

Let $3\left(x + \frac{4}{3}\right)^2 = \frac{2}{3}\tan^2 u \Rightarrow \sqrt{3}\left(x + \frac{4}{3}\right) = \frac{\sqrt{2}}{\sqrt{3}}\tan u \Rightarrow x = -\frac{4}{3} + \frac{\sqrt{2}}{3}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{2}}{3}\sec^2 u \Rightarrow \mathrm{d}x = \frac{\sqrt{2}}{3}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{4}{3x^2+8x+6}\mathrm{d}x$

$= \int \frac{4}{3\left(x+\frac{4}{3}\right)^2+\frac{2}{3}}\mathrm{d}x$

$= \int \frac{4}{\frac{2}{3}\tan^2 u+\frac{2}{3}} \times \frac{\sqrt{2}}{3}\sec^2 u\,\mathrm{d}u$

$= \int \frac{12}{2(\tan^2 u+1)} \times \frac{\sqrt{2}}{3}\sec^2 u\,\mathrm{d}u$

$= 2\sqrt{2}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int 2\sqrt{2}\,\mathrm{d}u$

$= 2\sqrt{2}u + c$

$= 2\sqrt{2}\arctan\left(\frac{x+\frac{4}{3}}{\frac{\sqrt{2}}{3}}\right) + c$

$= 2\sqrt{2}\arctan\left(\frac{3x+4}{\sqrt{2}}\right) + c$

So $x = -\frac{4}{3} + \frac{\sqrt{2}}{3}\tan u$ gives $\int \frac{4}{3x^2+8x+6}\mathrm{d}x = 2\sqrt{2}\arctan\left(\frac{3x+4}{\sqrt{2}}\right) + c$

5) f)

First: $5x^2 + 2x + 12 = 5\left(x + \frac{1}{5}\right)^2 + \frac{59}{5}$

Let $5\left(x + \frac{1}{5}\right)^2 = \frac{59}{5}\sinh^2 u \Rightarrow \sqrt{5}\left(x + \frac{1}{5}\right) = \frac{\sqrt{59}}{\sqrt{5}}\sinh u \Rightarrow x = -\frac{1}{5} + \frac{\sqrt{59}}{5}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{59}}{5}\cosh u \Rightarrow \mathrm{d}x = \frac{\sqrt{59}}{5}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{10}{\sqrt{5x^2+2x+12}}\mathrm{d}x$

$= \int \frac{10}{\sqrt{5\left(x+\frac{1}{5}\right)^2+\frac{59}{5}}}\mathrm{d}x$

$= \int \frac{10}{\sqrt{\frac{59}{5}\sinh^2 u+\frac{59}{5}}} \times \frac{\sqrt{59}}{5}\cosh u\,\mathrm{d}u$

$= \int \frac{10}{\sqrt{\frac{59}{5}(\sinh^2 u+1)}} \times \frac{\sqrt{59}}{5}\cosh u\,\mathrm{d}u$

$= \frac{10\times\frac{\sqrt{59}}{5}}{\sqrt{\frac{59}{5}}}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= 2\sqrt{5}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= 2\sqrt{5}\int 1\,\mathrm{d}u$

$= 2\sqrt{5}u + c$

$= 2\sqrt{5}\,\mathrm{arsinh}\left(\frac{5x+1}{\sqrt{59}}\right) + c$

So $x = -\frac{1}{5} + \frac{\sqrt{59}}{5}\sinh u$ gives $\int \frac{10}{\sqrt{5x^2+2x+12}}\mathrm{d}x = 2\sqrt{5}\,\mathrm{arsinh}\left(\frac{5x+1}{\sqrt{59}}\right) + c$

6) a)

Let $(2x-1)^2 = 5\sin^2 u \Rightarrow 2x-1 = \sqrt{5}\sin u \Rightarrow x = \frac{1}{2} + \frac{\sqrt{5}}{2}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{5}}{2}\cos u \Rightarrow \mathrm{d}x = \frac{\sqrt{5}}{2}\cos u\,\mathrm{d}u$

When $x = 0$, $u = \arcsin\left(-\frac{\sqrt{5}}{5}\right)$

When $x = 1$, $u = \arcsin\left(\frac{\sqrt{5}}{5}\right)$

$\therefore \int_0^1 \frac{2}{\sqrt{5-(2x-1)^2}}\mathrm{d}x$

$= \int_{\arcsin\left(-\frac{\sqrt{5}}{5}\right)}^{\arcsin\left(\frac{\sqrt{5}}{5}\right)} \frac{2}{\sqrt{5-5\sin^2 u}} \times \frac{\sqrt{5}}{2}\cos u\,\mathrm{d}u$

$= \int_{\arcsin\left(-\frac{\sqrt{5}}{5}\right)}^{\arcsin\left(\frac{\sqrt{5}}{5}\right)} \frac{2}{\sqrt{5(1-\sin^2 u)}} \times \frac{\sqrt{5}}{2}\cos u\,\mathrm{d}u$

$= \frac{2\sqrt{5}}{2\sqrt{5}} \int_{\arcsin\left(-\frac{\sqrt{5}}{5}\right)}^{\arcsin\left(\frac{\sqrt{5}}{5}\right)} \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \int_{\arcsin\left(-\frac{\sqrt{5}}{5}\right)}^{\arcsin\left(\frac{\sqrt{5}}{5}\right)} u\,\mathrm{d}u$

$= [u]_{\arcsin\left(-\frac{\sqrt{5}}{5}\right)}^{\arcsin\left(\frac{\sqrt{5}}{5}\right)}$

$= \arcsin\left(\frac{\sqrt{5}}{5}\right) - \arcsin\left(-\frac{\sqrt{5}}{5}\right)$

$= \arcsin\left(\frac{\sqrt{5}}{5}\right) + \arcsin\left(\frac{\sqrt{5}}{5}\right)$

So $x = \frac{1}{2} + \frac{\sqrt{5}}{2}\sin u$ gives $\int_0^1 \frac{2}{\sqrt{5-(2x-1)^2}}\mathrm{d}x = 2\arcsin\left(\frac{\sqrt{5}}{5}\right)$

6) b)

Let $(x-1)^2 = 4\sinh^2 u \Rightarrow x - 1 = 2\sinh u \Rightarrow x = 1 + 2\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\cosh u \Rightarrow \mathrm{d}x = 2\cosh u\,\mathrm{d}u$

When $x = 0$, $u = -\operatorname{arsinh}\left(\frac{1}{2}\right)$

When $x = 2$, $u = \operatorname{arsinh}\left(\frac{1}{2}\right)$

$\therefore \int_0^2 \frac{8}{\sqrt{4+(x-1)^2}}\mathrm{d}x$

$= \int_{-\operatorname{arsinh}\left(\frac{1}{2}\right)}^{\operatorname{arsinh}\left(\frac{1}{2}\right)} \frac{8}{\sqrt{4+4\sin^2 u}} \times 2\cosh u\,\mathrm{d}u$

$= \int_{-\operatorname{arsinh}\left(\frac{1}{2}\right)}^{\operatorname{arsinh}\left(\frac{1}{2}\right)} \frac{8}{\sqrt{4(1+\sinh^2 u)}} \times 2\cosh u\,\mathrm{d}u$

$= 8\int_{-\operatorname{arsinh}\left(\frac{1}{2}\right)}^{\operatorname{arsinh}\left(\frac{1}{2}\right)} \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= 8\int_{-\operatorname{arsinh}\left(\frac{1}{2}\right)}^{\operatorname{arsinh}\left(\frac{1}{2}\right)} u\,\mathrm{d}u$

$= [8u]_{-\operatorname{arsinh}\left(\frac{1}{2}\right)}^{\operatorname{arsinh}\left(\frac{1}{2}\right)}$

$= 8\operatorname{arsinh}\left(\frac{1}{2}\right) + 8\operatorname{arsinh}\left(\frac{1}{2}\right)$

$= 16\operatorname{arsinh}\left(\frac{1}{2}\right) = 16\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right) = 8\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)^2 = 8\ln\left(\frac{3+\sqrt{5}}{2}\right)$

So $x = 1 + 2\sinh u$ gives $\int_0^2 \frac{8}{\sqrt{4+(x-1)^2}}\mathrm{d}x = 8\ln\left(\frac{3+\sqrt{5}}{2}\right)$

6) c)

First: $45 + 24x + 4x^2 = 9 + 4(x+3)^2$

Let $4(x+3)^2 = 9\tan^2 u \Rightarrow 2(x+3) = 3\tan u \Rightarrow x = -3 + \frac{3}{2}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\sec^2 u \Rightarrow \mathrm{d}x = \frac{3}{2}\sec^2 u\,\mathrm{d}u$

When $x = -3$, $u = \arctan 0 = 0$

When $x = -\frac{3}{2}$, $u = \arctan 1 = \frac{\pi}{4}$

$\therefore \int_{-3}^{-\frac{3}{2}} \frac{3}{45+24x+4x^2}\,\mathrm{d}x$

$= \int_{-3}^{-\frac{3}{2}} \frac{3}{9+4(x+3)^2}\,\mathrm{d}x$

$= \int_{0}^{\frac{\pi}{4}} \frac{3}{9+9\tan^2 u} \times \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$= \int_{0}^{\frac{\pi}{4}} \frac{3}{9(1+\tan^2 u)} \times \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$= \frac{9}{18}\int_{0}^{\frac{\pi}{4}} \frac{\sec^2 u}{\sec^2 u}\,\mathrm{d}u$

$= \frac{9}{18}\int_{0}^{\frac{\pi}{4}} 1\,\mathrm{d}u$

$= \left[\frac{1}{2}u\right]_0^{\frac{\pi}{4}}$

$= \frac{1}{2}\left(\frac{\pi}{4}\right) - \frac{1}{2}(0) = \frac{\pi}{8}$

So $x = -3 + \frac{3}{2}\tan u$ gives $\int_{-3}^{-\frac{3}{2}} \frac{3}{45+24x+4x^2}\,\mathrm{d}x = \frac{\pi}{8}$

6) d)

First: $4x^2 - 12x + 15 = 6 + 4\left(x - \frac{3}{2}\right)^2$

Let $4\left(x - \frac{3}{2}\right)^2 = 6\sinh^2 u \Rightarrow 2\left(x - \frac{3}{2}\right) = \sqrt{6}\sinh u \Rightarrow x = \frac{3}{2} + \frac{\sqrt{6}}{2}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{6}}{2}\cosh u \Rightarrow \mathrm{d}x = \frac{\sqrt{6}}{2}\cosh u\,\mathrm{d}u$

When $x = 0$, $u = \operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)$

When $x = \frac{3}{2}$, $u = \operatorname{arsinh} 0 = 0$

$\therefore \int_0^{\frac{3}{2}} \frac{2}{\sqrt{4x^2-12x+15}}\,\mathrm{d}x$

$= \int_0^{\frac{3}{2}} \frac{2}{\sqrt{6+4\left(x-\frac{3}{2}\right)^2}}\,\mathrm{d}x$

$= \int_{\operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)}^{0} \frac{2}{\sqrt{6+6\sinh^2 u}} \times \frac{\sqrt{6}}{2}\cosh u\,\mathrm{d}u$

$= \int_{\operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)}^{0} \frac{2}{\sqrt{6(1+\sinh^2 u)}} \times \frac{\sqrt{6}}{2}\cosh u\,\mathrm{d}u$

$= \frac{2\sqrt{6}}{2\sqrt{6}} \int_{\operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)}^{0} \frac{\cosh u}{\sqrt{\cosh^2 u}}\,\mathrm{d}u$

$= \int_{\operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)}^{0} 1\,\mathrm{d}u$

$= [u]_{\operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)}^{0}$

$= (0) - \operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)$

$= -\ln\left(-\frac{\sqrt{3}}{\sqrt{2}} + \sqrt{\left(-\frac{\sqrt{3}}{\sqrt{2}}\right)^2 + 1}\right)$

$= -\ln\left(-\frac{\sqrt{3}}{\sqrt{2}} + \frac{\sqrt{5}}{\sqrt{2}}\right)$

$= -\ln\left(\frac{-\sqrt{3}+\sqrt{5}}{\sqrt{2}}\right)$

$= \ln\left(\frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}}\right)$

So $x = \frac{3}{2} + \frac{\sqrt{6}}{2}\sinh u$ gives $\int_0^{\frac{3}{2}} \frac{2}{\sqrt{4x^2-12x+15}}\,\mathrm{d}x = \ln\left(\frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}}\right)$

6) e)

First: $x^2 + 2x + 10 = 9 + (x+1)^2$

Let $(x+1)^2 = 9\sinh^2 u \Rightarrow x + 1 = 3\sinh u \Rightarrow x = -1 + 3\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\cosh u \Rightarrow \mathrm{d}x = 3\cosh u\,\mathrm{d}u$

When $x = -1$, $u = 0$

When $x = \frac{5}{4}$, $u = \operatorname{arsinh}\left(\frac{3}{4}\right) = \ln\left(\frac{3}{4} + \frac{5}{4}\right) = \ln 2$

$\therefore I = \int_{-1}^{\frac{5}{4}} \sqrt{x^2 + 2x + 10}\,\mathrm{d}x$

$= \int_{-1}^{\frac{5}{4}} \sqrt{9 + (x+1)^2}\,\mathrm{d}x$

$= \int_0^{\ln 2} \sqrt{9 + 9\sinh^2 u} \times 3\cosh u\,\mathrm{d}u$

$= \int_0^{\ln 2} \sqrt{9(1 + \sinh^2 u)} \times 3\cosh u\,\mathrm{d}u$

$= 9\int_0^{\ln 2} \sqrt{\cosh^2 u}\cosh u\,\mathrm{d}u$

$= 9\int_0^{\ln 2} \cosh^2 u\,\mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A \equiv \frac{1}{2}\cosh 2A + \frac{1}{2}$

$I = 9\int_0^{\ln 2} \left(\frac{1}{2}\cosh 2A + \frac{1}{2}\right)\mathrm{d}u$

$= 9\left[\frac{1}{4}\sinh 2A + \frac{1}{2}A\right]_0^{\ln 2}$

$= 9\left(\frac{1}{4}\sinh\big(2(\ln 2)\big) + \frac{1}{2}\ln 2 - \frac{1}{4}\sinh(0) - 0\right)$

$= \frac{9}{4}\sinh(\ln 4) + \frac{9}{2}\ln 2$

$= \frac{9}{4} \times \frac{1}{2}\left(\mathrm{e}^{\ln 4} - \mathrm{e}^{-\ln 4}\right) + \frac{9}{2}\ln 2$

$= \frac{9}{4} \times \frac{1}{2}\left(\mathrm{e}^{\ln 4} - \mathrm{e}^{-\ln 4}\right)$

$= \frac{9}{8}\left(4 - \frac{1}{4}\right) + \frac{9}{2}\ln 2$

$= \frac{135}{32} + \frac{9}{2}\ln 2$

So $x = -1 + 3\sinh u$ gives $\int_{-1}^{\frac{5}{4}} \sqrt{x^2 + 2x + 10}\,\mathrm{d}x = \frac{135}{32} + \frac{9}{2}\ln 2$

7)

First: $3 + 2x - x^2 = 4 - (x-1)^2$

Let $(x-1)^2 = 4\sin^2 u \Rightarrow x - 1 = 2\sin u \Rightarrow x = 1 + 2\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\cos u \Rightarrow \mathrm{d}x = 2\cos u\,\mathrm{d}u$

$\therefore \int \frac{5}{\sqrt{3+2x-x^2}}\mathrm{d}x$

$= \int \frac{5}{\sqrt{4-(x-1)^2}}\mathrm{d}x$

$= \int \frac{5}{\sqrt{4-4\sin^2 u}} \times 2\cos u\,\mathrm{d}u$

$= \int \frac{5}{\sqrt{4(1-\sin^2 u)}} \times 2\cos u\,\mathrm{d}u$

$= 5\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= 5\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int 5\,\mathrm{d}u$

$= 5u + c$

$= 5\arcsin\left(\frac{x-1}{2}\right) + c$

So that

$\int_0^1 \frac{5}{\sqrt{3+2x-x^2}}\mathrm{d}x = \left[5\arcsin\left(\frac{x-1}{2}\right)\right]_0^1 = 5\arcsin 0 - 5\arcsin\left(-\frac{1}{2}\right) = \frac{5\pi}{6}$ units^2

8)

First: $6 + 4x + x^2 = (x+2)^2 + 2$

Let $(x+2)^2 = 2\tan^2 u \Rightarrow x + 2 = \sqrt{2}\tan u \Rightarrow x = -2 + \sqrt{2}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sqrt{2}\sec^2 u \Rightarrow \mathrm{d}x = \sqrt{2}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{4}{6+4x+x^2}\mathrm{d}x$

$= \int \frac{4}{(x+2)^2+2}\mathrm{d}x$

$= \int \frac{4}{2\tan^2 u+2} \times \sqrt{2}\sec^2 u\,\mathrm{d}u$

$= \int \frac{2}{\tan^2 u+1} \times \sqrt{2}\sec^2 u\,\mathrm{d}u$

$= 2\sqrt{2}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int 2\sqrt{2}\,\mathrm{d}u$

$= 2\sqrt{2}u + c$

$= 2\sqrt{2}\arctan\left(\frac{x+2}{\sqrt{2}}\right) + c$

So $\int \frac{4}{6+4x+x^2}\mathrm{d}x = 2\sqrt{2}\arctan\left(\frac{x+2}{\sqrt{2}}\right) + c$

So that

$\int_{\sqrt{2}-2}^{\sqrt{6}-2}\left(x + \frac{4}{6+4x+x^2}\right)\mathrm{d}x = \left[\frac{1}{2}x^2 + 2\sqrt{2}\arctan\left(\frac{x+2}{\sqrt{2}}\right)\right]_{\sqrt{2}-2}^{\sqrt{6}-2}$

$= \left(\frac{1}{2}\left(\sqrt{6}-2\right)^2 + 2\sqrt{2}\arctan\left(\frac{\sqrt{6}-2+2}{\sqrt{2}}\right)\right) - \left(\frac{1}{2}\left(\sqrt{2}-2\right)^2 + 2\sqrt{2}\arctan\left(\frac{\sqrt{2}-2+2}{\sqrt{2}}\right)\right)$

$= \left(5 - 2\sqrt{6} + 2\sqrt{2}\arctan\sqrt{3}\right) - \left(3 - 2\sqrt{2} + 2\sqrt{2}\arctan 1\right)$

$= \left(5 - 2\sqrt{6} + 2\sqrt{2} \times \frac{\pi}{3}\right) - \left(3 - 2\sqrt{2} + 2\sqrt{2} \times \frac{\pi}{4}\right)$

$= 2 + 2\sqrt{2} - 2\sqrt{6} + \left(\frac{2}{3}\sqrt{2} - \frac{1}{2}\sqrt{2}\right)\pi$

$= \left(2 + 2\sqrt{2} - 2\sqrt{6} + \frac{\sqrt{2}}{6}\pi\right)$ units^2